

# Things I Don't Understand About Quantum Error Correction

A naive audit of the constructs · CWI Summer School on QA & QEC · 24–28 August 2026 · Poster Session Wed 26 Aug

## The Thesis (one paragraph)

Every human construct for understanding quantum computing is **internally consistent** — and the imports **contradict each other**. Spiders (categorical algebra), Pauli webs (stabilizer/QEC), and gadgets (MBQC/compilation) are maps drawn by different cartographers: coherent in the computational domain, contradictory where 3+1D particle physics and 1D thermodynamics are mixed into a 2D drawing plane without a compatibility check. *The diagrams are not wrong — they are maps. The discipline of using a map is knowing where it ends.* (Anchor: "ZX Diagrams at the Seam", 10.5281/zenodo.21992118)

## The Method: feigned naivete as probe

I learned the room's language (the Bruhat–Tits tree  $T_2$  as a ZX graph state — fig4) — **and that is when the seams became visible**. I ask the questions I'm not supposed to need to ask. The naive question is the external structural probe: where imports contradict, understanding breaks — and **that break is data, not deficit**.

## Things I don't understand about QEC

1. **Why the discreteness?** Is it physical, or imported from the circuit model? If the abstraction is a choice, fault tolerance is the tax on the choice.
2. **What is a logical qubit, physically?** Hardware is bosonic; codes assume qubits. The map is 2D; the machine is continuous.
3. **What does a threshold actually guarantee?** Proofs assume stochastic local noise; coherent/correlated errors are unpriced seams.
4. **Where is the quantum in the loop?** Syndrome → classical decoder → feedback. QEC is quantum memory + classical control.
5. **Why isn't the decoder part of the code?** A classical algorithm decides a quantum code's performance — two silos, never audited together.
6. **Why 2D geometry?** Why a surface code and not a tree? What if error topology is ultrametric — the staircase nobody has looked for?
7. **Why count T gates, not joules?** The field prices the algebraic resource; nobody prices the ancilla factory, the retries, the decoder.
8. **Why is the cat the unit of protection?** Shor states are cat states — exponentially fragile, verified ancillas go stale. Protection from something that fragile?

## The statement & the honest caption

**The one-line statement:** "I cannot grasp quantum computing through the practitioners' constructs because the constructs are maps drawn by different cartographers — internally consistent, mutually contradictory. **My confusions are the fault lines.**"

**Honest caption:** feigned naivete is a method, not a pose — the naive question finds the seam. I am not claiming the field is wrong; I am claiming the imports were never checked against each other. *If the constructs were compatible, the naive questions would answer themselves.*

## Questions for the room

1. "Is the discreteness of the circuit model physical or imported?" · 2. "Has anyone priced the energy of the decoder — per correct solution?" · 3. "Why a 2D cluster and not a tree?"

**References:** ZX Diagrams at the Seam — 10.5281/zenodo.21992118 (concept 21991895) · Universal Ignorance Audit — 10.5281/zenodo.21901984 · Knowing What We Do Not Know — 10.5281/zenodo.21901983 · Ultrametric Code Spaces — 10.5281/zenodo.21824194 · JPCUB — 10.5281/zenodo.21945415 · arXiv:2601.04467 · arXiv:2410.17992 · arXiv:2012.13966 · arXiv:1902.03178

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